Highly accurate protein structure prediction with AlphaFold

Paper by Jumper et al., *Nature* 596 (2021)

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Machine Learning for Genomics (263-5351-00L), ETH Zürich, Fall 2026

Outline

1. The problem
2. The benchmark
3. The network
   The data
   The network, in detail
4. Training
5. What makes it work
6. Conclusion
7. Questions

The problem: sequence to structure

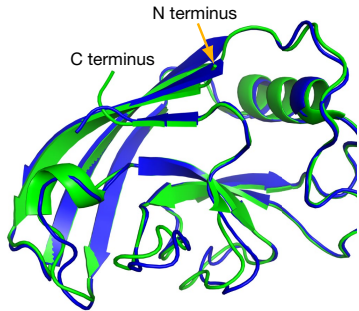
SIFSYITESTGTPS…

sequence of amino acids. Each has the same backbone and its own side chain

Structure experiments take months to years

Hundreds of millions of sequences are known

Predict 3D atomic coordinates from the sequence alone



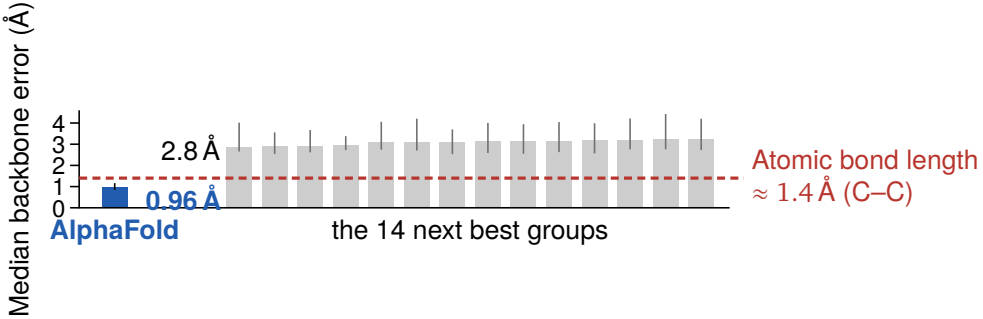
3D coordinates of all heavy atoms
Prediction in blue, experiment in green

[JEP+21] Jumper et al., *Nature* 2021, Fig. 1b. Target T1049 (PDB 6Y4F), CASP14.

The benchmark: CASP14

CASP14 (May to July 2020): blind test, structures withheld until submission

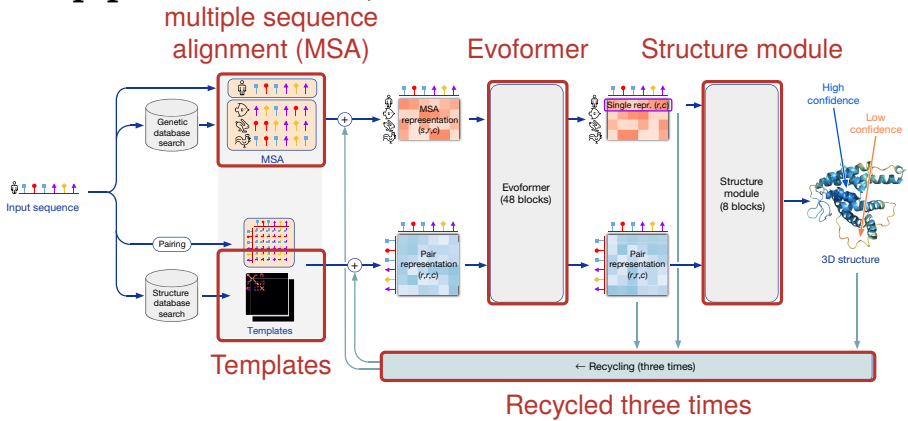
Metric: Cα root-mean-square deviation on the 95% best-model residues



The median backbone error is 0.96 Å—below one carbon atom width

Redrawn from Fig. 1a. Cα r.m.s.d.$_{95}$ on 87 domains (95% CIs). 1 Å = 10^{-10} m.

The pipeline: MSA in, coordinates out



The network outputs the 3D coordinates directly

[JEP+21] Jumper et al., *Nature* 2021, Fig. 1e.

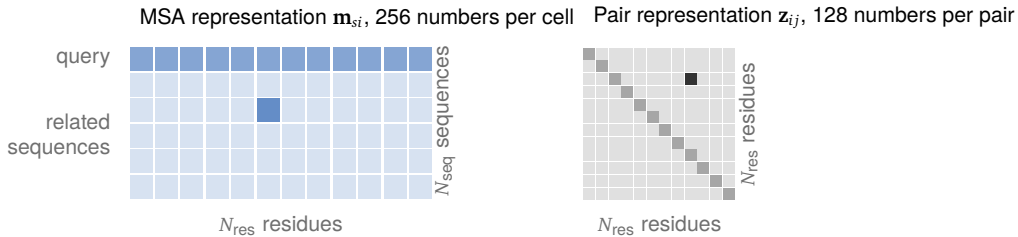
Four facts, four design choices

		Fact	Design choice in AlphaFold 2
1		An MSA is a set of sequences. A protein is a chain of residues.	Attention over the sequences themselves, along the rows and the columns.
2		Distances in 3D must close triangles.	The pairs are the edges of a residue graph, updated through triangles.
3		A structure is the same after any rotation or translation.	One frame per residue. Attention uses distances; updates in local frames.
4		A protein and its mirror image are different molecules.	The loss compares atoms in local frames. A mirror image is penalised.

Removing any one of the four costs accuracy. The ablations measure how much.

The two representations

Two tables of numbers, one per MSA cell and one per residue pair

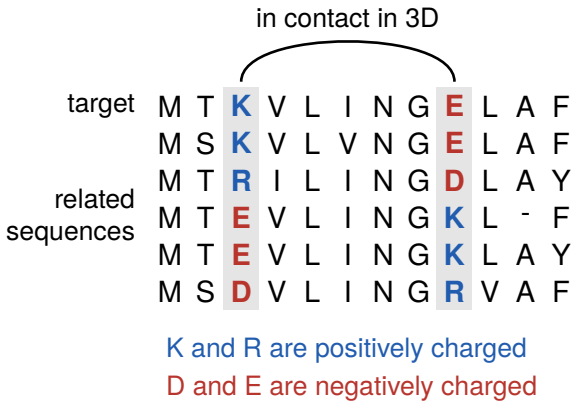


One vector per cell of the alignment, and one per ordered pair of residues

Each of the 48 Evoformer blocks reads and writes only these two arrays

Supplementary Information (SI) 1.5, Algorithms 3 and 4. Sizes in training: 128 sequences, crops of 256 residues (SI Table 4).

The input: MSA representation



A row is a related protein

A column is an aligned position

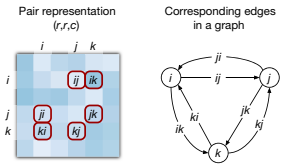
Correlated mutations reveal contacts in 3D

The network reads the raw alignment, not its summary statistics

[MHC+11] D. S. Marks et al., *PLoS ONE* 6, 2011. The alignment is an illustration, not data.

The input: Pair representation

What z_{ij} holds and where it comes from



128 numbers for every ordered pair of residues, the edge $i \rightarrow j$ of the graph

Started from amino acid pairs and chain distance; templates enter as distances

At the end, the distogram reads a distance distribution for every pair from it

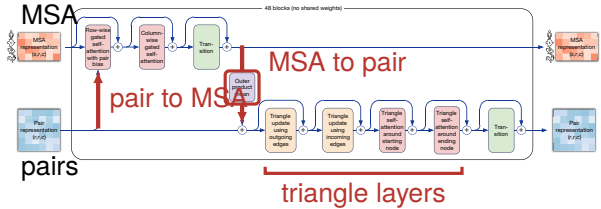
Fig. 3b, the pairs as directed edges

The pair array stores distances before any coordinates exist

SI 1.5, Algorithm 4 (relative position), 1.7.1 (templates), 1.10 (recycling), 1.9.8 (distogram).

The Evoformer block

1. The sequences in an MSA are a set. The residues are a chain.



Row attention: residues attend to residues, biased by the pair edge $\mathbf{z}_{ij}$

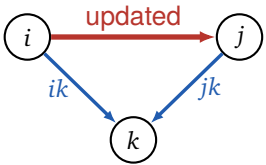
Column attention across sequences; outer product mean adds coevolution into pairs

The paper calls this graph inference: pairs steer the MSA, MSA updates the pairs

[JEP+21] Jumper et al., *Nature* 2021, Fig. 3a, 48 blocks. Algorithms 7 to 10. Detailed figures in backup 11 and 13.

Evoformer: triangle updates

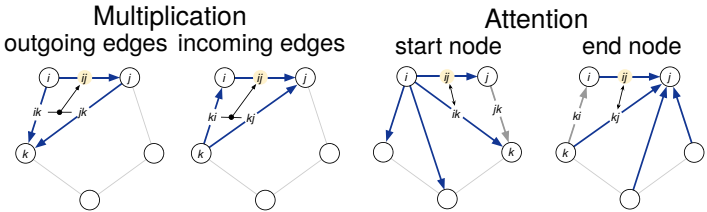
2. Distances in 3D must close triangles.



The pair representation forms the edge set of a complete residue graph

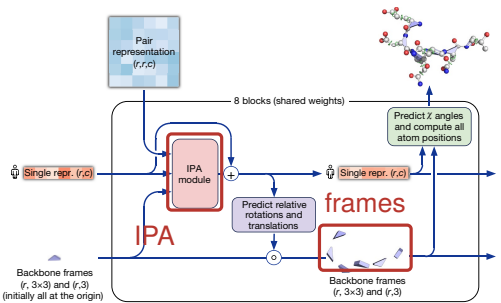
Edge (i, j) is updated by summing over third nodes k: (i, k) and (j, k)

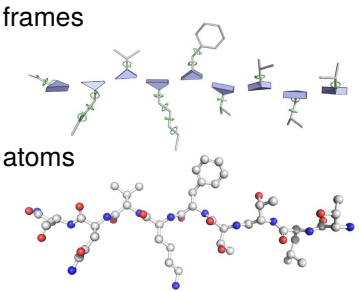
Triangle updates enforce metric spatial consistency directly in pair space



Structure module: residues as rigid bodies

3. A structure is the same after any rotation or translation.





A frame is one rotation and one translation per residue: SE(3)

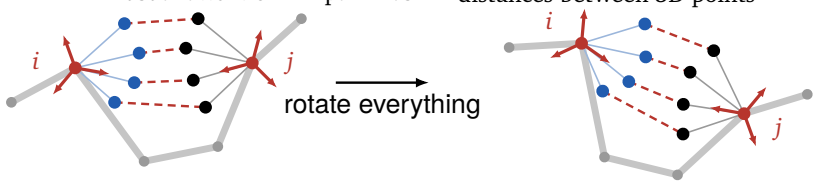
IPA updates residue states; all frames move simultaneously in local coordinates

Residues move as an unconstrained “residue gas”; peptide bonds are learned

Invariant point attention

3. A structure is the same after any rotation or translation.

$$a_{ij} = \text{softmax}_j\Big(\underbrace{\boldsymbol{q}_i^\top \boldsymbol{k}_j}_{\text{usual attention}} + \underbrace{b_{ij}}_{\text{pair bias}} - \underbrace{\textstyle\sum_p \left\| T_i \circ \vec{\boldsymbol{q}}_{ip} - T_j \circ \vec{\boldsymbol{k}}_{jp} \right\|^2}_{\text{distances between 3D points}} \Big)$$



Points are placed in the frames of i and j

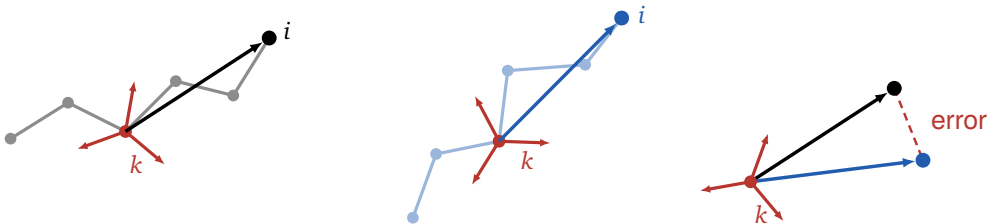
The distances stay the same

Attention weights and scalar updates are strictly $\mathrm{SE}(3)$-invariant

The formula is Algorithm 22 of the supplement for one head, without scaling constants.

Frame aligned point error (FAPE)

4. A protein and its mirror image are different molecules.



experiment prediction both, seen from frame k

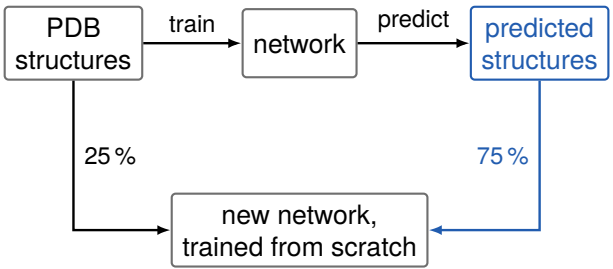
$$\text{FAPE} = \frac{1}{Z} \underset{k,i}{\text{mean}} \min\left(d_{\text{clamp}}, \left\|(\boldsymbol{R}_k, \boldsymbol{t}_k)^{-1}\boldsymbol{x}_i - (\boldsymbol{R}_k^{\text{true}}, \boldsymbol{t}_k^{\text{true}})^{-1}\boldsymbol{x}_i^{\text{true}}\right\|\right)$$

A rotation of the prediction changes nothing. A mirror image is penalised.

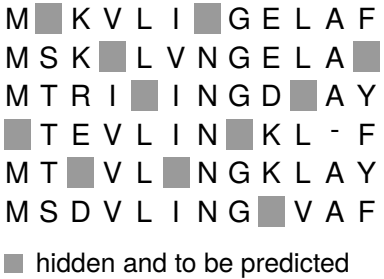
Supplementary Information, Algorithm 28, with $Z = d_{\text{clamp}} = 10\,\text{Å}$. The mirror case is on backup 3.

Training data: PDB and self-distillation

Self-distillation adds 350,000 predicted structures



Masked-MSA loss trains on unlabelled sequences

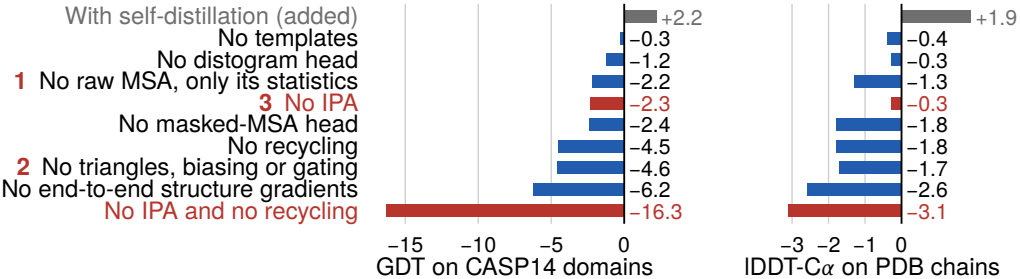


Taps into uncharacterised sequence space via self-distillation and masking

[XLHL20] Xie et al., CVPR 2020 (self-distillation). [DCLT19] Devlin et al., NAACL 2019 (BERT masking).

Ablations: the cost of each component

1 2 3 4

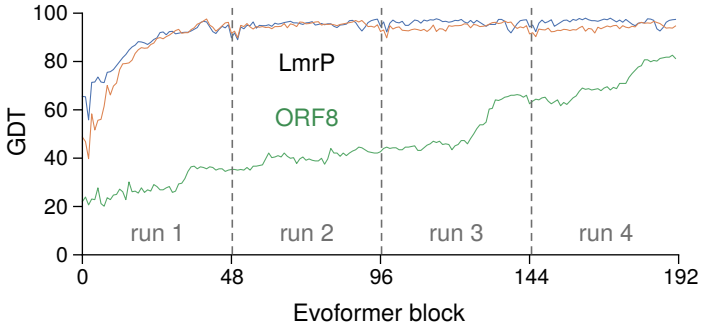


4 Without FAPE, a mirror-blind loss often builds the mirror image (Suppl. Fig. 9)

Each of the four choices costs accuracy on CASP14 when removed

Redrawn from Fig. 4a, mean change vs baseline over training runs. GDT and lDDT 0–100. Row 3 (No IPA) removes iterative refinement but keeps frames and FAPE (SI 1.13.1). Line 4: Suppl. Fig. 9.

Intermediate structures in the trunk

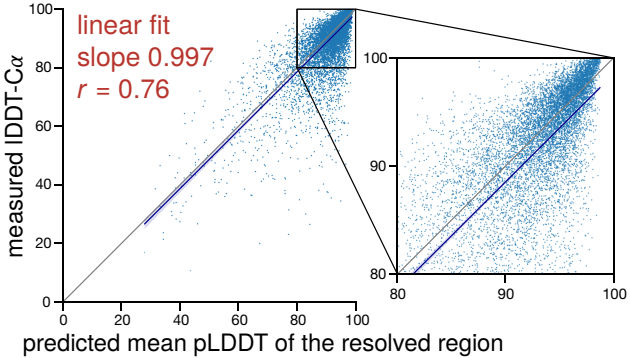


The trunk is frozen and every block gets a structure module

LmrP is correct after about 25 blocks, ORF8 only after about 150

[JEP+21] Jumper et al., *Nature* 2021, Fig. 4b. LmrP: transporter (PDB 6T1Z); ORF8: SARS-CoV-2 (T1064).

Confidence: pLDDT against measured lDDT



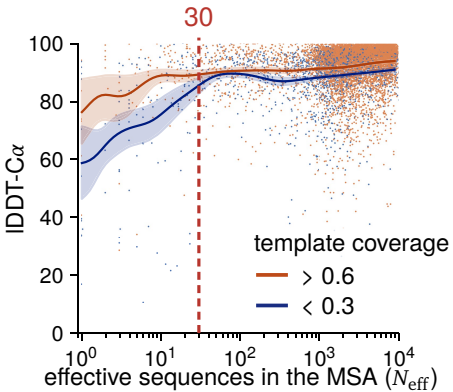
pLDDT predicts local Cα frame accuracy per residue

Low pLDDT (< 50) flags intrinsically disordered regions

Calibrated confidence enables filtering without ground truth

[JEP+21] Jumper et al., *Nature* 2021, Fig. 2c, n = 10,795 chains. Following [TAW+21] Tunyasuvunakool et al., *Nature* 2021.

Limitations: MSA depth and complexes



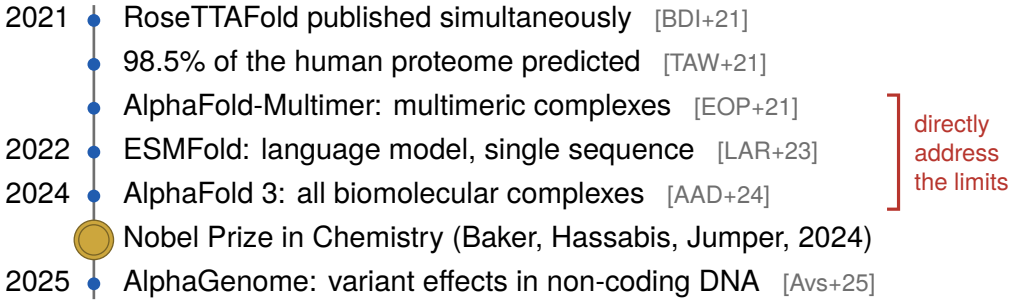
Accuracy drops sharply when $N_{\text{eff}} < 30$

Chains shaped by complexes have higher error

AlphaFold 2 predicts static, isolated monomers

[JEP+21] Jumper et al., *Nature* 2021, Fig. 5a.

After AlphaFold 2: 2021 to 2025



AlphaFold 3 generalizes across biomolecular complexes, outperforming specialized tools

Exception: AIchemy_RNA2 on CASP15 RNA targets. Complete citations in backup References.

AlphaFold 3: which choices survived

Abramson et al. [AAD+24], *Nature* 630 (2024), pages 493 to 498

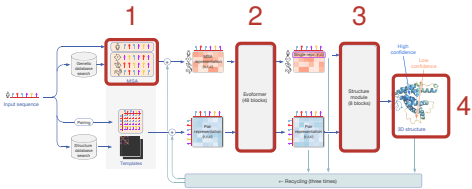
	AlphaFold 2	AlphaFold 3 verdict	The cost / patch
1	raw MSA attention	small MSA block, not kept	same MSA depth dependence
2	triangles on the pairs	kept unchanged, 48 blocks	inductive bias remains key
3	frames, IPA, equivariance	dropped for diffusion	clashes; ranking and distillation
4	FAPE in local frames	dropped; aligned MSE loss	4.4 % chirality errors; ranking, seeds

Unlike AlphaFold 2, AlphaFold 3 provides no component ablation study

The bitter lesson, with caveats: the graph survived; chirality required post-hoc ranking

[Sut19] R. Sutton, *The Bitter Lesson*, 2019. AF3 reports 4.4% chirality violations and relies on multi-seed ranking. Architecture: backup 14.

Summary: four design choices, measured



		GDT lost
1	attention over sequences	−2.2
2	triangles on the pairs	−4.6
3	frames and IPA	−2.3
4	FAPE in local frames	mirrored

Trained end to end. One prediction takes about a GPU minute

Four facts became four design choices. Each costs accuracy when removed.

[JEP+21] Jumper et al., *Nature* 2021. Figures reproduced under CC BY 4.0. CASP14: 0.96 Å median backbone error.

Exam question

Machine Learning for Genomics (263-5351-00L)

Official Assignment

Assigned by Prof. Valentina Boeva and course instructors 7 days prior to presentation date.

[Instructor question prompt inserted here]

Structured Solution

1. Inductive bias and architecture
2. Mathematical loss formulation
3. Genomic & biological impact

Rigorous theoretical justification and step-by-step derivation

Questions for the audience

1 AlphaFold 2 hardcoded SE(3) equivariance and stereochemical chirality into architecture and loss. AlphaFold 3 replaced both with unconstrained diffusion. When is geometric inductive bias indispensable, and when is unconstrained generative modeling preferable?

2 Accuracy falls below about 30 related sequences. Has the network learned how proteins fold, or how evolution has varied them? What result would settle it?

3 pLDDT is trained on the model's own errors. On which proteins would you distrust it, and what should a genomics pipeline do with a low-confidence region?

Why is IPA invariant to rigid motions?

SI 1.8.2, Algorithm 22, equations 3 to 6



The only geometry in the weights is the distance between points that i and j place in their own frames

$$\left\|(T_{\text{global}} \circ T_i) \circ \vec{\boldsymbol{q}}_i - (T_{\text{global}} \circ T_j) \circ \vec{\boldsymbol{k}}_j\right\|^2$$
$$= \left\|T_{\text{global}} \circ (T_i \circ \vec{\boldsymbol{q}}_i - T_j \circ \vec{\boldsymbol{k}}_j)\right\|^2 = \left\|T_i \circ \vec{\boldsymbol{q}}_i - T_j \circ \vec{\boldsymbol{k}}_j\right\|^2$$

moving both points together leaves their distance unchanged

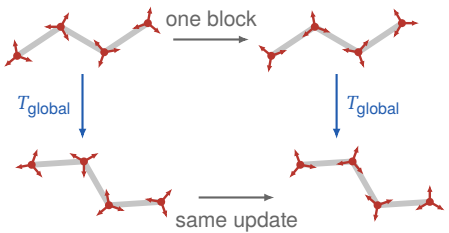
$$(T_{\text{global}} \circ T_i)^{-1} \circ \textstyle\sum_j a_{ij}\, (T_{\text{global}} \circ T_j) \circ \vec{\boldsymbol{v}}_j$$
$$= T_i^{-1} \circ T_{\text{global}}^{-1} \circ T_{\text{global}} \circ \textstyle\sum_j a_{ij}\, T_j \circ \vec{\boldsymbol{v}}_j = T_i^{-1} \circ \textstyle\sum_j a_{ij}\, T_j \circ \vec{\boldsymbol{v}}_j$$

the output points are read back in frame i, and T_{global} meets its inverse. Exact, not learned.

Weights and outputs do not depend on where the structure sits

Why is the structure module equivariant?

SI 1.8.2 and 1.8.3, Algorithms 20 and 23



The update of a frame is predicted from $\boldsymbol{s}_i$, which is invariant, and composed in the frame's own coordinates

$(T_{\text{global}} \circ T_i) \circ T_i^{\text{upd}} = T_{\text{global}} \circ (T_i \circ T_i^{\text{upd}})$

moving the input first or the output last gives the same frames

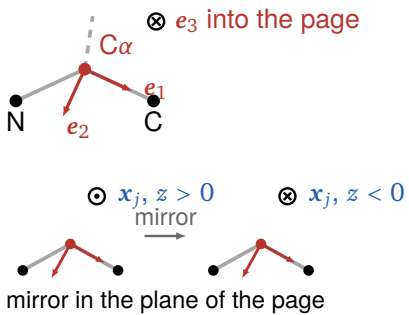
"Rigid motion" in the supplement means proper rotation and translation: SE(3).

Frames initialize as the identity; the loss must be invariant to the initial pose

Exact SE(3) equivariance. Chirality is enforced by FAPE and chiral data.

Origin of stereochemical chirality

SI 1.8.1, 1.9.3 and 1.8.4, Algorithm 21, equations 11 to 17



A frame is built from N, Cα and C by Gram–Schmidt, with $\boldsymbol{e}_3 = \boldsymbol{e}_1 \times \boldsymbol{e}_2$. It is always right-handed.

Reflect all atoms. The frames do not reflect with them. In local coordinates exactly z flips:

$$T_i^{-1} \circ \boldsymbol{x}_j \;\mapsto\; \mathrm{diag}(1, 1, -1)\,(T_i^{-1} \circ \boldsymbol{x}_j)$$

$$\mathrm{FAPE}(X, X_{\mathrm{mirror}}) = 2 \sum_{ij} \left|(T_i^{-1} \circ \boldsymbol{x}_j)_z\right|$$

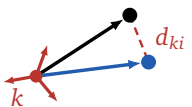
large unless the structure is flat

Trained with dRMSD, the error of the distance matrix, which is blind to reflection, the model builds the mirror image in a large share of cases (Suppl. Fig. 9)

FAPE breaks reflection symmetry; PDB training data teach L-amino acid chirality.

What does FAPE measure?

SI 1.9.2, 1.9.4 and 1.9.5, Algorithm 28, equations 18 to 30



seen from frame k

Every atom i in every frame k, in local coordinates: $N_{\text{frames}} \times N_{\text{atoms}}$ terms, each clamped at 10 Å and divided by 10 Å

Invariant to a rigid motion of the prediction, and of the truth, by the same cancellation as in IPA

Lower bound: FAPE is at least the RMSD after the best rigid alignment, whatever the frames (eqs. 18 to 22). So it is zero only if the two point sets differ by a proper rigid motion.

Non-negative, symmetric, triangle inequality: a pseudometric on structures (eqs. 23 to 30)

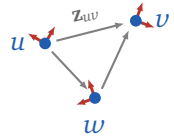
Local geometry in every frame: blind to placement, sensitive to the hand

Is AlphaFold 2 a graph neural network?

Formalism of Bronstein, Bruna, Cohen and Veličković, 2021

$$\mathbf{h}_u = \phi\Big(\mathbf{x}_u, \bigoplus_{v \in \mathcal{N}_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{e}_{uv})\Big)$$

message ψ, aggregation $\bigoplus$, update ϕ. Nodes: residues, $\mathbf{s}_u$ and frame T_u. Edges: $\mathbf{z}_{uv}$.



Block	message ψ	into
MSA row attention	$\psi(\mathbf{m}_{su}, \mathbf{m}_{sv}, \mathbf{z}_{uv})$, the edge biases the weight	node u
outer product mean	$\psi(\mathbf{m}_{\cdot u}, \mathbf{m}_{\cdot v})$, averaged over the sequences	edge uv
triangle updates	$\psi(\mathbf{z}_{uv}, \mathbf{z}_{uw}, \mathbf{z}_{vw})$ over third nodes w (out and in)	edge uv
IPA	$\psi(\mathbf{s}_u, \mathbf{s}_v, \mathbf{z}_{uv}, \lVert T_u \circ \vec{\mathbf{q}}_u - T_v \circ \vec{\mathbf{k}}_v \rVert)$, values moved by $T_u^{-1} \circ T_v$	node u
backbone update	none, ϕ alone	frame T_u

Triangles combine two edges into the third to enforce 3D spatial consistency.

Geometry enters through invariants: distances in weights, relative frames in values.

What does each block compute?

Supplement, Algorithms 6 to 32

	Update	Message	SI
	Evoformer, 48 blocks		
1	$\mathbf{m} \leftarrow \mathbf{m} + \text{RowAttention}(\mathbf{m} \mid \mathbf{z})$	node → node, edge biases the weight	Alg. 7
2	$\mathbf{m} \leftarrow \mathbf{m} + \text{ColumnAttention}(\mathbf{m})$	across the sequences at one node	Alg. 8
3	$\mathbf{m} \leftarrow \mathbf{m} + \text{MLP}(\mathbf{m})$	per node	Alg. 9
4	$\mathbf{z} \leftarrow \mathbf{z} + \text{OuterProductMean}(\mathbf{m})$	node pair → edge	Alg. 10
5	$\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleMultiplication}(\mathbf{z})$, out and in	edge pair → edge	Alg. 11, 12
6	$\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleAttention}(\mathbf{z})$, start and end node	edge pair → edge	Alg. 13, 14
7	$\mathbf{z} \leftarrow \mathbf{z} + \text{MLP}(\mathbf{z})$	per edge	Alg. 15
	Structure module, 8 blocks with shared weights. At the start $\mathbf{s}_u = \mathbf{m}_{1u}$ and T_u is the identity.		
8	$\mathbf{s} \leftarrow \mathbf{s} + \text{IPA}(\mathbf{s}, \mathbf{z}, T)$	node → node, values moved by $T_u^{-1} \circ T_v$	Alg. 22
9	$\mathbf{s} \leftarrow \mathbf{s} + \text{MLP}(\mathbf{s})$	per node	Alg. 20
10	$T_u \leftarrow T_u \circ \text{BackboneUpdate}(\mathbf{s}_u)$	per node, in its own frame	Alg. 23
	Recycling, 4 cycles		
11	$\mathbf{z} \leftarrow \mathbf{z} + \text{Embed}(d_{uv})$, C$\beta$ distances from the frames	frames → edges	Alg. 32

Every line is $\mathbf{h}_u = \phi(\mathbf{x}_u, \bigoplus_v \psi)$ on nodes or on edges. Information moves in lines 1, 4, 5, 6, 8 and 11.

What exactly did each ablation remove?

SI 1.13.1 to 1.13.3

Row of Fig. 4a	What the model has instead
No raw MSA	the profile and the pairwise frequencies of the MSA, in one row, no column attention
No triangles, biasing or gating	axial attention on the pairs with a relative position bias, no gating
No IPA	frames from one linear projection of the single representation, no iteration
No recycling	a single pass through the network
No end-to-end structure gradients	the structure losses stop before the Evoformer, which learns from the auxiliary losses only
No templates	no template stack, four more MSA clusters instead

Baseline: the full model without self-distillation. Three or four seeds, not re-tuned

Not additive: no IPA costs 2.3, no recycling 4.5, both together 16.3 GDT

How is confidence predicted?

SI 1.9.6 and 1.9.7, Algorithm 29

pLDDT: a small network on the final single representation predicts, per residue, a distribution over 50 bins of lDDT-Cα. It is trained with cross-entropy against the lDDT-Cα of the prediction itself, on structures resolved between 0.1 and 3 Å

The reported pLDDT is the expected value of that distribution. The pLDDT of a chain is the mean over its residues

pTM, the predicted template modelling score: a pairwise head predicts the distribution of the aligned error of every residue pair, and the TM-score follows from these distributions

Confidence is a prediction, trained on the model's own errors

How was it trained?

Methods, “Training regimen”, SI 1.3 and 1.11.1, Table 4

Data: PDB structures released before 30 April 2018, and about 350,000 predicted structures of Uniclust30 sequences without a structure, filtered by confidence, 75 % of the samples

Stage 1: crops of 256 residues, about 10 million samples, about seven days on 128 TPU v3 cores

Stage 2: crops of 384 residues, the structural violation loss switched on, about 1.5 million samples, about four days

Losses: FAPE, torsion angles, distogram, masked MSA, pLDDT, experimentally resolved, violations

Eleven days on 128 TPU cores per model. Inference takes GPU minutes to hours.

AlphaFold 3: further changes

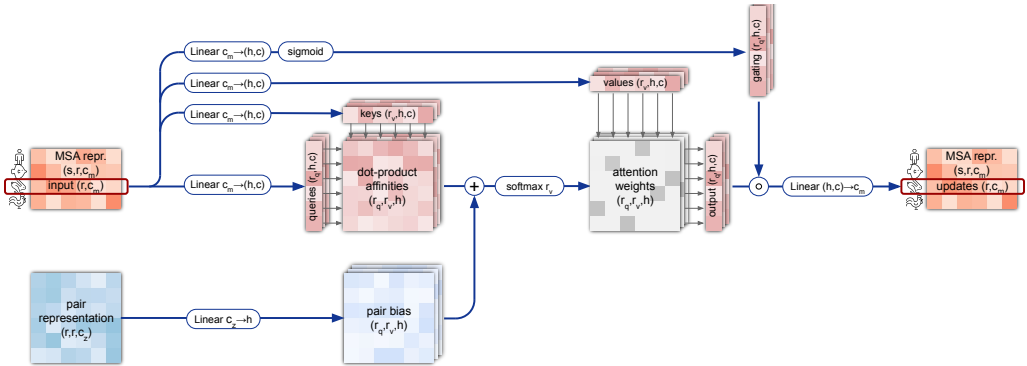
Abramson et al. [AAD+24], *Nature* 630 (2024)

Input	tokens, one per residue or per ligand atom	p. 493
Trunk	pairformer, 48 blocks. MSA block of 4, MSA not kept	p. 494
Output	diffusion over raw atoms, no frames, random rotations instead	p. 495, SI Alg. 19
Confidence	trained on 20-step mini rollouts	p. 495
Sampling	5 seeds, 5 samples each, ranked with clash and chirality penalties	p. 494, 497
Limits	chirality 4.4 %, clashes, hallucination, no dynamics	p. 496, 497

Less structure in the network, more in the data and in the sampling

MSA row attention with pair bias

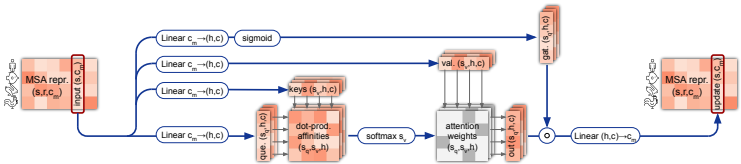
Supplementary Fig. 2, Algorithm 7



Supplementary Fig. 2. Every sequence of the MSA attends along itself, the pair representation biases the weights

Evoformer: columns and outer product mean

Supplementary Figs. 3 to 5, Algorithms 8 to 10



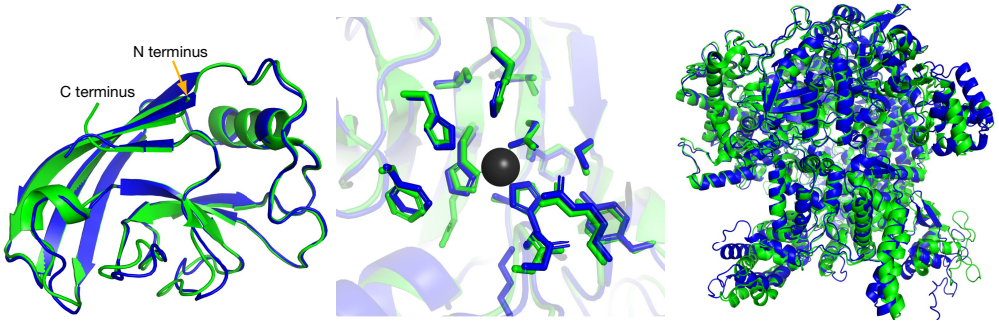
One residue at a time: the sequences attend to each other at position *i*, then a two-layer perceptron per cell

Outer product mean: the outer product of columns *i* and *j*, averaged over the sequences, is added to $\mathbf{z}_{ij}$

Here the alignment writes into the pairs, in every one of the 48 blocks

Prediction and experiment superposed

CASP14 targets: backbone, side chains and large chains (Fig. 1b to d)

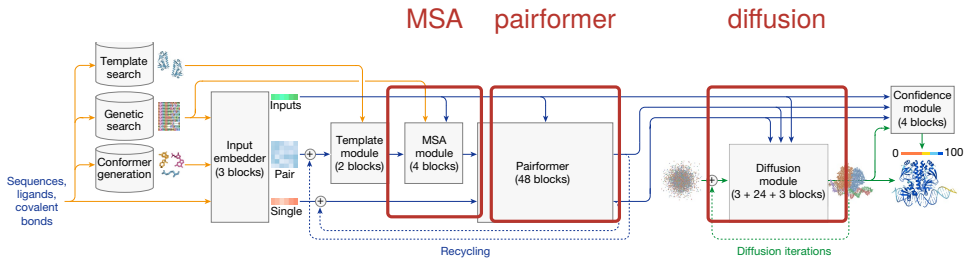


AlphaFold experiment

Prediction and experiment overlap, also in the side chains

The AlphaFold 3 architecture

Abramson et al. [AAD+24], *Nature* 630 (2024), Fig. 1d



Tokens for proteins, nucleic acids, ligands and ions

Diffusion denoises random atom coordinates, 25 samples ranked by confidence

Less structure in the network, more in the data and in the sampling

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